COMPOSITIONAL IMPLICATIONS OF TOUCH SCREEN TECHNOLOGY

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ABSTRACT

Different ways of performing live improvisatory computer music using touch screen technologies are discussed and examples of performing with these are shown. Performing with these screens give new performing and composing possibilities, and these are discussed, along with compositional details of works made with this technology.

1. XENAKIS AND GABURO

As early as 1952, in his diary, Iannis Xenakis wrote "Relearning to touch sound with our hands – that's the heart, the essence of music!"[1] and his lifelong involvement with creating music through the hand ("The hand is the organ of the body that is closest to the brain.")[2] culminated in the UPIC system, where waveforms, gestures and musical forms could be specified on a drawing tablet.

Kenneth Gaburo's work of the 1980s involved him in an intense use of physical processes and sensory deprivation experiments as a means of generating musical material for his compositions. He maintained that his body would be a better, and more intrinsic, source of quasi-random information than any external source could be. His works Antiphony IX (orchestra, children and electronics) and Re-run (electronics) were both generated with sensory-deprivation processes and these are described in his papers ISIT, and LA.[3]

[1] Makhi Xenakis, *Meteorites*, in "Iannis Xenakis: Architecct, Composer, Visionary", p 130, The Drawing Center, New York, 2010, issuu.com/drawingcenter/docs/drawingpapers88_xenakis (last accessed April 20, 2014)

[2] ibid, Sharon Kanach, *"Music to be seen: Tracing Xenakis's Creative Process"*, p.99

[3] Kenneth Gaburo, *CONCERNING PHYSICALITY: Harry Partch's BEWITCHED*;(aesthetic philosophic discourse); Percussive Arts Magazine (Research Edition), V.23, #3; March, 1985
Kenneth Gaburo, *ISIT* (a philosophical argument with reference to the concept of subject-object; also (synchronously), a text sound-performance composition); Pub. International Synergy Journal, V. 1, #2; Los Angeles; 1987 http://www.angelfire.com/mn/gaburo/isit.html
Kenneth Gaburo, *LA*: (a philosophical argument for Sound As Spirit); Pub. Perspectives of New Music; V. 25, #1; 1987

2. A DILEMMA AND AN INTEREST

I've become dissatisfied with small physical controllers and netbooks – although they are very convenient, they limit the physicality of one's performance to a very small area, and they don't allow for whole body engagement in performance, some things I've been searching for for several decades now. I've also become dissatisfied (or maybe that's frustrated due to lack of access to them and time to work with them) with controllers that simply read the position of the body in space, something I had worked with extensively in the 1980s and 90s, such as the 3DIS system and the Buchla Lightning.[4] (Warren Burt "Fair Exchanges: *Writings on Dance* No. 18-19,Melbourne, Winter 1999, pp. 51-71. Warren Burt: "Computer as Part of Improvisatory Theatrical Performance. Or What I Did with Eva and Bill in July and August 2002." in *Chroma* No. 32, Brisbane, August 2002, pp. 9- 14.)

Most recently I've become intrigued with the possibilities of touch-screen technology. It's relatively cheap, easily accessible, and it seems to offer a number of possibilities for whole-body performance, as well as offering potentials for compositional resources I haven't encountered before. Whenever I encounter a new piece of equipment or software, my first question is "What are the compositional possibilities of this?" and "How can it extend what I already do or offer me new possibilities?"

For the past couple of years I've been investigating what I could do with touch-screen technology. I've done this on 3 platforms – an Apple iPad4, an ASUS Vivo Tab Windows 8 tablet/netbook, and a Samsung Galaxy Tab2 7.0 Android Tablet. Additionally, I've used a series of used iPhone and Android mobile phones as auxiliary sound-makers in performances. This paper will detail some of the pieces I've made with this technology and what some of the implications of this technology are.

[4] Warren Burt "Fair Exchanges: *Writings on Dance* No. 18-19,Melbourne, Winter 1999, pp. 51-71.
Warren Burt: "Computer as Part of Improvisatory Theatrical Performance. Or What I Did with Eva and Bill in July and August 2002." in *Chroma* No. 32, Brisbane, August 2002, pp. 9- 14.)

3. SEVERAL KINDS OF INTERACTION

3.1. Drawing

Drawing – there are a number of programs which allow one to draw either spectra or pitch/time contours, such as HighC, Coagula, and Virtual ANS. Already, like Xenakis, I had found working with a drawing tablet instead of a mouse was very useful with these programs. Now with touch-screen technology, even just a single-touch screen, I find I can make spectrograph drawings and draw pitch/time contours with my fingers. Immediately after I started this I realized that I was doing a kind of sonic fingerpainting. I found the musical gestures I was making with my hand were very different from the kinds of gestures I had been inscribing with a mouse. They featured more elaborate shapes that curved back on themselves with more facility than any mouse movement could. Here is an example of a pitch/time score made with HighC.

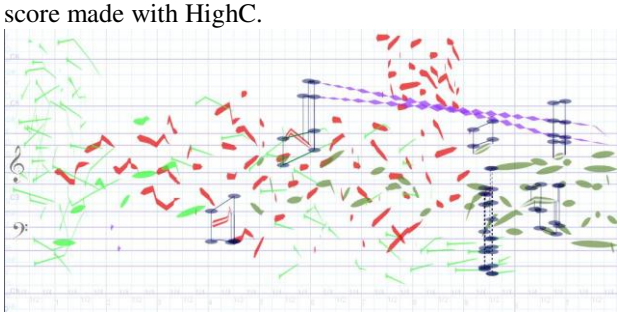


Figure 1: Warren Burt: Ides of March score made with HighC.

I made a composition with drawings like this over several days – it's called "Audio Fingerpainting." It's the first section of what I hope will be a much longer work. http://www.warrenburt.com/storage/AudioFingerpaintingMix.mp3

3.2. Multiple Tablets / Multiple Apps

For the launch of Linda Kouvaras' book "Loading the Silence: Australian Sound Art in the Post-Digital Age"[5], I was asked to give a speech / do a small performance. This performance took place in the Grainger Museum of the University of Melbourne. Surrounded by exhibits of Percy's experiments with Free Music, it occurred to me that I could do a performance with multiple tablets which would have sound sources that would reflect various aspects of Grainger's (and others') experiments. I had 5 different tablets. The Windows8 tablet had Ross Bencina's AudioMulch on it, which was running a very simple program that played pre-programmed glides of sine waves, emulating Grainger's 1950s Free Music experiments. All I had to do was touch "start" at the proper time, and the Windows8 tablet would play a short gliding tone sequence as an homage. I had two Android tablets, both of which were running the John Cage Prepared Piano App. Since Cage was working on his prepared piano work simultaneously with Grainger's Free Music experiments, I thought this would be appropriate. Further, having the app on two tablets at once meant I could have a kind of polyphony and virtuosic-sounding gesture that wouldn't have been possible with just one tablet. Finally, on the iPad and an iPhone, I was running the Thumbjam app. On the iPhone I had a sampled Grand Piano tuned to 36 tones per octave equal temperament, like Grainger's Butterfly Piano experimental instrument. On the iPad, I had Thumbjam playing a series of samples drawn from Grainger's own recordings of his Butterfly Piano experiments. Having samples of these allowed me to insert Percy's actual gestures into the mix.[6]

Having five tablets allowed me to play as if I had an extended percussion sampler available. The aspects of single-hits and brief visiting of particular sound worlds encouraged me to leap rapidly between the tablets, making a kind of live fantasy that I don't think I would have been able to make in any other way. Here's a short excerpt from a video of a performance made after the launching.



Figure 2: Warren Burt: Launching Piece, June 2013 http://www.warrenburt.com/journal/2013/6/21/launching-piece-composed-and-performed-by-warren-burt-a-new.html (view 3:00-4:00)

3.3. Microtonal Keyboard Work

A number of apps have come along which allow one to set up microtonal keyboards of various kinds, some of which allow one to route MIDI from their outputs to other synthesizers. Some of these programs are SoundSquares, Musix Pro, and Lemur, which offers 2 options: one can either program one's own experimental keyboard arrays, or use their new Hexapads template. In all these cases, one can set up two dimensional arrays of either rectangular or hexagonal keys which allow one different harmonic and performance possibilities than a traditional keyboard can. For example, in "Morning on Princes Pier," I use the SoundSquares app to play a series of 7-note scales derived from Ervin Wilson's "The Scales of Mt Meru" series of papers.[7] I route the output

[5] Linda Kouvaras, *"Loading the Silence: Australian Sound Art in the Post-Digital Age"*Ashgate, London, 2013

[6] "Picking Up The Threads of a History More Extensive Than Previously Known: Percy Grainger's Work with Music Technology." - in "*Proceedings of the 2004 ACMC Conference*", ACMA, Wellington.

[7] http://www.anaphoria.com/wilsonintroMERU.html (last accessed April 20, 2014)

of the iPad into an Alesis Air F/X, which allows me to move my hand above the unit, controlling the effects. My left hand performs pitches, and my right hand controls effects. By setting the two units a good distance apart, I can have wide-reaching stance, which will allow me much more freedom of movement than performing on one device would. Notice here that I've bought a stand which allows me to put the iPad on a mic stand and perform with it in a free standing manner. This allows me to have an upright posture, which I think will be a more engaging way of performing than sitting down will. That is, by standing, I can move around more, and in a more expressive manner, than I could if I were seated. This is more engaging for me, and I hope also for an audience.



Figure 3: Warren Burt: Morning on Princes Pier 2013 for iPad and Alesis AirF/X
http://www.warrenburt.com/journal/2013/5/13/concert-at-box-hill-institute-may-9-2013.html (view 7:20-8:20)

In "Lucas (2-1) C Right Drone" I assemble a custom rectangular keyboard in Lemur which allows me to play different 7 and 5 note subsets of a particular microtonal scale I derived in 2006 from Ervin Wilsons "Scales of Mt Meru" paper. Different "pages" of the Lemur keyboard allow me to freely switch between different ways of dividing up the scale. Further, there are two parallel keyboards for each harmonic division of the scale: One consists of toggle switches which turn on and off sustained pitches, while the other consists of momentary switches with only produce sound for the duration of touch, as a normal keyboard would. With these parallel keyboards I can assemble complexes of sustained tones as well as melodic filigree above them. And I can freely "modulate" between different harmonic worlds which are present in my 12-note microtonal scale. In this piece, the custom keyboard is on the iPad in the Lemur app. The MIDI from the iPad is routed into a Windows computer, which is running LinPlug's Spectral softsynth. More recently, I've also routed the MIDI from the Lemur keyboard internally on the iPad to the Sunrizer softsynth on the iPad, a way of performing this piece with only one piece of gear.

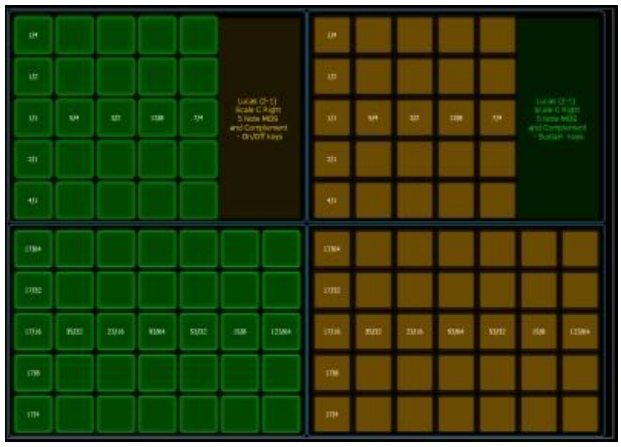


Figure 4: Lemur Keyboard layout on iPad for "Lucas (2-1) C Right Drone" Warren Burt 2014 http://youtu.be/JPi9iqz_Hvk (view 0:10-1:10)

Most recently, I've been working with Hexagonal keyboard arrangements on the iPad using both Musix Pro as well as with the Hexapad Layouts in Lemur. (I was just about to sit down for several months and program my own hexagonal keyboard program, and then not one, but two programs to do that showed up within weeks of each other. This has happened several times in the past few years. There are so many people now developing things that it's almost as if you want something, all you have to do is wait a while, and an app to do it will be developed by someone, somewhere.) Both programs allow some re-configuring of the keyboards, and I've been finding that applying these "generalized keyboard" layouts to non-equal tempered microtonal scales has been helping me to find harmonic resources in microtonal scales and making them easy to play in a way that even working with a physical hexagonal keyboard (such as the Microzone keyboard) wouldn't. The Microzone, at least the one that the University of Wollongong had, was very hard to play – the keys were physically hard to press down – after 15 minutes playing it, I had very sore wrists. The hexagonal layouts on the iPad on the other hand, allow me to glide my fingers easily over the surface, triggering off arpeggios and harmonic riffs that at the moment are a source of delight. Here is a video example of a short live piece "Tenebrae Chorale" in which the Lemur Hexagonal Keyboard (using two different keyboard arrangements) controls the Sunrizer synth playing in an 11 note microtonal MOS scale as developed by Ervin Wilson. This performance takes place using only the iPad.

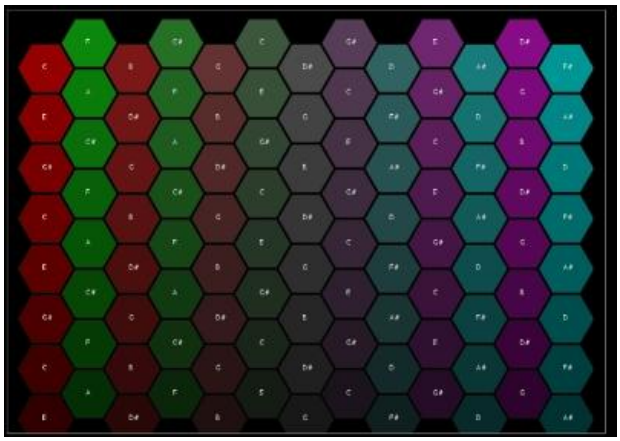


Figure 5: Hexagonal Keyboard Layout in Lemur on iPad http://youtu.be/HM4KstUcr_E (view 0:15-1:15)

3.4 **"Keyboard" layouts that allow live performance and direct control of waveforms themselves, or control of aspects of a process.**

A number of apps allow one to directly perform on a representation of the sound waveform, such as MegaCurtis and Samplr, while others such as TopLap and abcdefgh...allow one control of only semi-predictable processes. For example, Samplr allows you to use up to 6 different samples, each of which can have their waveforms manipulated in real-time with a variety of slicing, looping, and scrubbing techniques. In this piece, "A Degenerate Practice", a quick video demonstration reveals some of the software instrument's capabilities. Six loops are used – two are electronic soundscapes I fingerpainted in Henry Lowengard's iOS based "Tondo" app, another is a loop of music by the late Brazilian accordionist and composer Sivuca. Other sources include a loop from Robert Erickson's "East of the Beach," the opening chords of Beethoven's "Waldstein Sonata," and a quote from American composer Harry Partch.



Figure 6: iPad screen layout for "A Degenerate Practice" http://youtu.be/OtiQQ3jZCVw (view 1:00-2:15)

A number of apps use the accelerometer in the iPad to give a sense of "gravity" to the events on the screen. One of the most clever of these is Jorg Piringer's abcdefghij.....in which samples of German-language phonemes bounce around the screen, being played when they hit a boundary of the screen. The movement of the letter-icons changes as the iPad is tilted. In the performance of my "Lettrist Fantasy" I pick alphabet beads at random from a small box. These provide the phonemes I load into Piringer's program, and the tilting of the iPad changes the patterns of the phonemes. I use the graphics of the program as a score to determine which phonemes I will sing/speak and how I will do that. The use of small belt-mounted loudspeakers in this piece gives me total freedom of movement, so that I'm not even constrained by a connection to a sound system. Also, the use of belt-mounted loudspeakers means that sound from both the electronics and my voice are coming from the same (movable) point in space.



Figure 7: Warren Burt performing Lettrist Fantasy, May 2013http://www.warrenburt.com/journal/2013/5/13/concert-at-box-hill-institute-may-9-2013.html (view 2:30-3:30)

3.5 Multiple Controls

These can be either traditional – like the sliders of a mixer – or they can be more radical – such as setting up a custom performance layout – or they can be a set of controls that simply allow one to navigate through the possibilities of an algorithmic process. For example, I set up a Lemur patch on the iPad to control a patch on a Windows8 computer, using John Dunn's ArtWonk and the Modartt Pianoteq physically modeled virtual piano, which made a canon from the output of one of Julien Sprott's chaotic equations.[8] The Lemur keyboard allows one to turn the patch on and off, select one of 8 tempi, work with 9 different transpositions of the 3 voices of the canon, set 4 different melodic widths for the Sprott equation generated melodies, select 8 different "modes" which use different pitches of the microtonal scale used in the piece, select 3 different

[8] Julien Sprott "Simple Chaotic Systems and Circuits." (Am. J. Phys. 68(8), August 2000, pp. 758-63

ranges for the volume of the initial melody (which is canonically imitated by the other two voices) and have 3 different settings of the pedals of the Pianoteq. With this amount of control, I can move through the space of compositional possibilities of the patch as rapidly or slowly as I wish. That's a lot of different kinds of structures which I can explore with this setup.

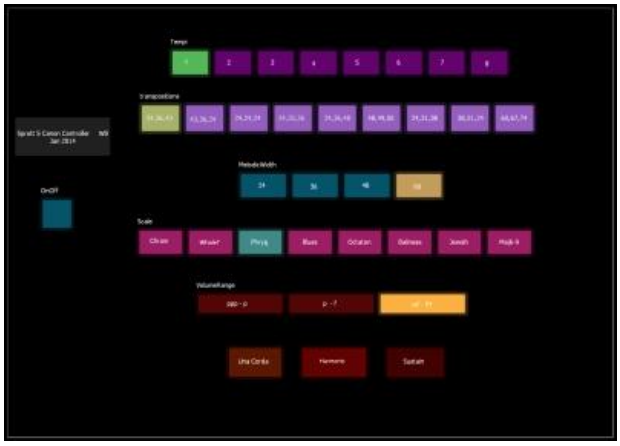


Figure 8: Lemur Keyboard Layout on iPad for "Sprott Canon Controller" 2014 Warren Burt http://youtu.be/lyTf0J35S2o (view 0:15-1:15)

Another kind of setup is a more direct performance setup – "Four Parisian Mods". In this one, the iPad Lemur patch provides 4 x-y controllers, each of which controls the parameters in 4 different GRM Tools plugins. The MIDI from the iPad is sent to a Windows8 PC, on which the GRM tools are hosted by AudioMulch. Auxiliary sliders in the Lemur patch also control if an effect is heard or not. With a mouse, one can control just one x-y window on one GRM Tool at a time, but with this setup, one can easily control 4 different effects simultaneously. Also, the attempt to meaningfully control 4 different x-y controllers at once on a touch screen leads to some fairly comical physical results.

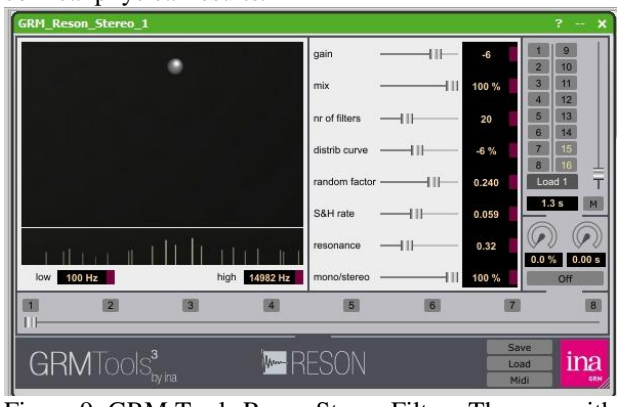


Figure 9: GRM Tools Reson Stereo Filter. The area with the ball in the upper left is the area where parameters of the plugin can be controlled with a mouse or an external controller, such as an iPad based Lemur patch.

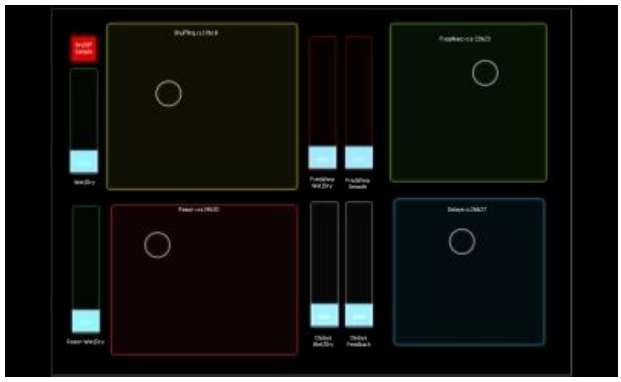


Figure 10: Lemur Control Surface on iPad for "4 Parisian Mods" Warren Burt 2014 http://youtu.be/c93pQzND5dA (view 1:00-2:00)

4. CONCLUSION

One of the essential things about designing a performing interface for a piece is that when you are doing that, you are designing the visual environment that you will be living in while performing. The importance of this cannot be overstressed. Given a visual setup, this will influence the way you perform the piece, and the more deeply you sink into the visual world of the interface, the more deeply you will become enmeshed in the compositional possibilities inherent in your musical material. Further, designing a visual interface, with musical performance possibilities in mind, will create a richer conceptual environment in which to move.

I think it's more interesting to watch someone perform with their full body rather than watching them sitting behind a laptop (even if that only involves someone standing behind a non-obscuring device instead of sitting). (My wife calls this the "Reading One's Email in Public" brand of theatre.) Further, the use of touch-screen technology, either with existing apps or custom made, allows one a greater range of gestures and compositional possibilities than other, more traditional modes of performing. Despite the unfortunate implications of the term, we actually DO want this kind of music making to have some aspects of "the spectacle"[9] about it, so that our performing WILL be an occasion "to be present to."[10] In the future, larger performance platforms might make the performance more physically engaged. For example, I think a 1 meter by 1 meter touch screen could allow for some wonderfully energetic and extravagant full body gestures. But for the moment, concentrating on a single screen, or several screens, will allow performance which both directly shapes individual gestures and also controls aspects of an algorithmic process, and does it quite elegantly.

[9] Guy Debord: *Society of the Spectacle* http://www.antiworld.se/project/references/texts/The_Society%20_Of%20_The%20_Spectacle.pdf

[10] Benjamin Boretz: If I Am a Musical Thinker, *Perspectives of New Music*, Vol 20, No. 1 & 2 (Autumn 1981 – Summer 1982) 464-517.

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